FPGA Parallelism – Lab 2

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HLS synthesized latency:

```
================================================================
== Performance Estimates
================================================================
+ Timing:
    * Summary:
    +--------+---------+----------+------------+
    |  Clock |  Target | Estimated| Uncertainty|
    +--------+---------+----------+------------+
    |ap_clk  |  10.00 ns|  7.300 ns|     2.70 ns|
    +--------+---------+----------+------------+

+ Latency:
    * Summary:
    +---------+---------+----------+----------+--------+--------+---------+
    |  Latency (cycles) |  Latency (absolute) |     Interval    | Pipeline|
    |   min   |   max   |    min   |    max   |  min   |  max   |   Type  |
    +---------+---------+----------+----------+--------+--------+---------+
    |   131667|   131667|  1.317 ms|  1.317 ms|  131668|  131668|       no|
    +---------+---------+----------+----------+--------+--------+---------+
```

Co-simulation latency and clock period:

RTL	Status	Latency(Clock Cycles) min	Latency(Clock Cycles) avg	Latency(Clock Cycles) max	Interval(Clock Cycles) min	Interval(Clock Cycles) avg	Interval(Clock Cycles) max	Total Execution Time (Clock Cycles)
VHDL	NA	NA	NA	NA	NA	NA	NA	NA
Verilog	Pass	116311	116311	116311	NA	NA	NA	116311

Post-implementation resource utilization:

```
#=== Post-Implementation Resource usage ===
SLICE:            0
LUT:          20697
FF:            9353
DSP:              0
BRAM:           321
URAM:             0
LATCH:            0
SRL:            106
CLB:           3735

#=== Final timing ===
CP required:                     10.000
CP achieved post-synthesis:      4.857
CP achieved post-implementation: 8.313
Timing met
```

Final speedup ratio calculation:

116311 cycles * 8.313 post-imp. clock period = 0.96689 ms

143.5702 / 0.96689 = <u>148.5702 speedup</u>